Multimodal Anemia Detection Using Nail and Eye Images with Numerical Parameters

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Abstract: ***One of the common blood disease is Anemia that creates difficulty for the body to carry oxygen because it lowers the levels of haemoglobin. Traditional methods of diagnosing anaemia requires invasive blood tests performed in laboratories by trained professionals, which can be time- consuming and painful for patients. In this work, we propose a more patient-friendly approach: a multimodal machine learning system that uses non-invasive images of the eye and nail, with blood parameters such as Mean Corpuscular Volume (MCV), Haemoglobin (Hb), Mean Corpuscular Haemoglobin (MCH), and gender. There are two branches of deep learning in the system. One branch, based on DenseNet169, collects features from pictures, and the other works with numbers. Final prediction is done by a fusion layer puts the two branches together. After being trained on a larger dataset with TensorFlow, the model got an accuracy of 94–95%. With OpenCV, you can click photos and make predictions right away to check for anaemia. It is quicker, less intrusive, and easier to get than regular tests. This approach is a huge step forward in helping clinicians diagnose anaemia more easily.***

Keywords: Anemia, Hemoglobin, DenseNet169, Deep Learning, Multimodal Data Fusion, Biomedical Image Processing.

I. INTRODUCTION

The World Health Organisation (WHO) [1] says that about 1.9 billion people areaffected by anaemia which is one of the most frequent blood diseases in the world.It happens when the amount of haemoglobin or red blood cells in the body lowers. People who have this disease could feel weary, pale, disoriented, short of breath, and unable to focus. If you have untreated anaemia, it can have serious consequences on your health, such causing issues during pregnancy and slowing down brain development. Diagnosing and treating anaemia early is very important to minimise these difficulties and improve quality of life. This is especially true for groups that are more likely to have it, like children, women of reproductive age, and the elderly. The most common ways to find out if someone has anaemia is by taking Blood tests that check for hemoglobin, mean corpuscular hemoglobin and mean corpuscular hemoglobin concentration, mean corpuscular volume [2]. These measures provide extensive clinical insights on the morphology and functionality of red blood cells (RBCs). But these exams have a lot of big difficulties. For example, they need blood samples from veins and practiced medical professionals, and they need laboratory set-up to do haematological analysis. It can take a long time for the findings to come in, sometimes even hours. The procedures are invasive, which makes them uncomfortable and less likely to be followed, especially for kids and older people. Access is limited in remote or low-resource locations where there aren't many labs. These issues demonstrate the necessity for an affordable, semi non-invasive, and rapid method to test for anaemia that may be employed in community health clinics, primary care units and remote rural locations.

Sometimes, the outward indications of anaemia show up as pale skin on the tongue, conjunctiva (eye), and nail bed [3]. Doctors commonly use these visual cues to make a first check. persons with anaemia have nails that are lighter in colour because they don't have as much haemoglobin [4]. Healthy persons have pinkish nails. The inner eyelid changes colour depending on how much oxygen-rich blood is going through it. The skin may look a little different or uneven when oxygen levels decline. Recent advancements in computer vision and digital image processing have enabled the objective recording and analysis of these visual indicators [5]. Now, instead of depending on their own valuation, clinicians may use deep learning algorithms to measure subtle changes in texture, brightness, and colour that are linked to the severity of anaemia. This change has made it practicable to diagnose without cameras or surgery, which has made healthcare easier to get and more computerized. Machine Learning (ML) and Deep Learning (DL) have changed the way doctors diagnose diseases by allowing computer models learn from data that comes from many different places. Convolutional neural networks (CNNs) can look at photos of people with anaemia to determine signs of pallor, vascularity, and tissue colour. Traditional models can look at numbers to figure out how dreadful the disease is. But single-modality systems (which only use image data or only numerical values) often have problems like unreliable lighting or camera conditions that make images less clear, differences in skin tone between people that cause unfairness, and a lack of

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complementary quantitative information that makes decisions less precise. Multimodal AI techniques use both visual and numerical data to demonstrate the external phenotypic characteristics and the internal haematological characteristics of anaemia. The proposed work presents a Multimodal Anaemia Detection System that combines image data from the nail bed and eye with quantitative haematological pointers to predict anaemia status. The system includes two parts that work together: the Image Branch, which uses DenseNet169, a pre-trained convolutional neural network architecture that is known for its dense connectivity and has the skill to reuse features [6]. Gets deep insights from pictures of nails and eyes that have to do with changes in colour intensity, brightness, and texture Numerical Branch: Uses neural layers to work with structured features like Haemoglobin (Hb), MCV, MCH, MCHC, and Gender, finding out how terrible anaemia is and how numbers are related to each other. The fusion layer integrates the outputs from both branches, employing the best aspects of image-based and numerical analysis to generate a final classification output of "Anaemic" or "Non-Anemic." This method lets the model make more precise and reliable diagnostic predictions. The system uses two main data sources to generate a powerful dataset: the Image Dataset, which has pictures of nails and eyes collected in varied lighting. The collection includes both real-time photos taken with OpenCV from live camera feeds and openly available medical image records [7]. Numerical Dataset: Has tables with blood test results. Each entry is for a patient and has one or more photographs attached to it. Image preprocessing involves making the image smaller (to 224×224 pixels), normalising the data, and adding more data by rotating, flipping, and zooming in. StandardScaler makes sure that all the features are scaled the same way by normalising numerical data [8]. The model is trained and evaluated with an 80:20 split to keep the class balance.

Python TensorFlow and tools like NumPy, Pandas, and Scikit-learn were used to create the model. The most crucial processes are: DenseNet169 layers trained on ImageNet take visual embeddings. Building a Fusion Network: The features from both modes are combined and sent through dense layers [9]. The Binary Cross Entropy loss and Adam optimizer work together to make sure that gradient descent is stable and converges swiftly. We use Accuracy, Precision, Recall, F1- score, and AUC (Area Under Curve) [10] to quantify how well something works. You may also get and forecast photos in real time with OpenCV integration. Users can type in numbers and view the results of the classification right immediately.

The purpose of this work is to build a hybrid multimodal architecture that can work in real time [11]. The proposed method diverges from traditional or unimodal AI models by employing both visual and numerical cues to reduce diagnostic ambiguity. It uses transfer learning with DenseNet169, which shortens the training time and improves feature generalisation [12]. It uses OpenCV-based live testing, which means it may be used in real life for point-of- care screening. It is a low-cost, semi-automated solution that works on regular desktops and laptops. This approach illustrates how the integration of clinical data analytics with image-based deep learning may facilitate the development of user-friendly diagnostic tools for primary healthcare and telemedicine [13].

II. Methodology

The Multimodal Anaemia Detection System was made to combine non-invasive imaging features with numerical haematological data into one AI framework [14]. The major goal was to build a diagnostic algorithm that could reliably tell if someone has anaemia and work in real time, which would mean fewer intrusive blood tests. Getting data, cleaning it up, making a model, learning through fusion, training and testing, and testing in real time are all included.

A. The structure of the system

The proposed system architecture is based on a two-branch deep learning model that consists of: Image Branch: A deep convolutional neural network (DenseNet169) that uses features from photos of nails and eyes.

Numerical Branch: A fully connected dense neural network that looks at blood parameters like Mean Corpuscular Volume (MCV), Hemoglobin (Hb), Mean Corpuscular Hemoglobin (MCH), Mean Corpuscular Hemoglobin Concentration (MCHC).

Fusion Layer: A layer that puts together the feature vectors. Classification Head: A last dense layer with a sigmoid activation function that puts people into two groups: Anaemic (1) or Non-Anemic (0).

The architecture based on fusion makes sure that the model learns both visual representations of the signs of anaemia and quantitative physiological data. This makes the predictions more trustworthy.

Figure 1 depicts how the proposed multimodal anaemia detection system will work in general. It illustrates the two branches of images and numbers that work together to make the final classification.

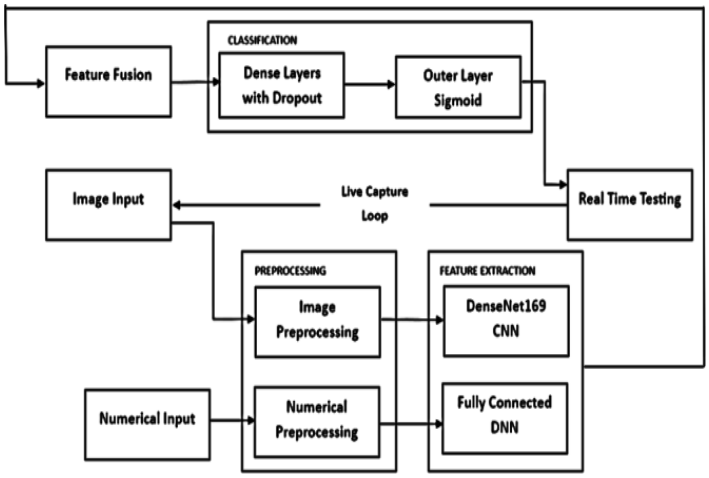


FIGURE 1: MULTIMODAL ANEMIA DETECTION SYSTEM ARCHITECTURE COMBINING IMAGE AND NUMERICAL FEATURE BRANCHES WITH FUSION-BASED CLASSIFICATION

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B. Description of the Dataset Image Collection

The image dataset was made to show signs of anaemia using pictures of nails and the conjunctiva (the part of the eye that covers the eyeball):

Source: Publicly accessible medical repositories augmented by photos obtained via OpenCV from volunteers in regulated lighting conditions.

Dataset size: The dataset contains around 1,000 to 1,200 images, with around half of them showing people who are Anaemic and half showing people who are not.

Resolution: Each image was produced to be 224 × 224 pixels in RGB format. Although the dataset is small, the transfer learning using DenseNet169 reduces data dependency by leveraging pre trained features from large datasets. And also data augmentation techniques were used to mitigate overfitting.

Tagging: Labels were assigned based on the levels of haemoglobin or clinical data that had been verified.

The photographs exhibit varying skin tones, lighting, and angles, which makes the model more diversified and better at making generalisations.

Numeric Dataset

The numerical dataset includes demographic and haematological data taken from Kaggle anemia related datasets. The dataset includes subjects of varying age groups and skin tones to improve robustness. The parameters are: Mean Corpuscular Volume (MCV) means the average size of red blood cells.

Hemoglobin (Hb) is the most important factor in how much oxygen can be carried.

Mean Corpuscular Hemoglobin (MCH) means the average amount of haemoglobin in each red blood cell.

Mean Corpuscular Hemoglobin Concentration (MCHC) means the average quantity of haemoglobin in a red blood cell. Gender - 1 represents male and 0 represents female.

By using the CSV dataset (which has one row for each image sample), the model learns to correlate visual and numerical data.

C. Data Preprocessing Image Preprocessing

Image preprocessing makes sure that everything is the same and gets rid of noise before the model takes it in.

Resizing: All photos were resized to 224×224×3 to meet the size of DenseNet169's input.

Normalisation: To speed up convergence, the pixel values were converted so that they were between 0 and 1. Augmentation: Keras's Image Data Generator was used to do things like random rotation (0°–40°), horizontal and vertical flipping, zoom range (0.2), and brightness variation (±10%) inorder to make the model stronger and stop it from overfitting.

Colour correction: Histogram equalisation was employed to remedy the colour problems caused by uneven lighting.

Numerical Data Preprocessing

We cleaned, normalised, and encoded the numerical data: Normalisation: StandardScaler from Scikit-learn made sure that all continuous values had a mean of zero and a unit variance.

Encoding: The gender was set to 0 for women and 1 for men. Data Integration: The CSV file linked each entry to the image path. The resulting output is a latent representation vector capturing relationships between hematological parameters and anemia status.

Fusion and Classification

The Concatenate layer in Keras put together the outputs from both branches. After the fusion, two more thick layers (128 and 64 neurones) were added to make learning joint features better. The final classification layer uses a sigmoid activation to yield a binary probability value (1 = Anaemic, 0 = Non-Anemic).

Mathematical Formulations of the Proposed Classifiers

1. *Image Branch: DenseNet169 Feature Extraction*

The input to the image branch is an RGB image of size 224 × 224, as defined in Eq. (1):

$$X \in \mathbb{R}^{224\times224\times3} \quad (1)$$

This image serves as the raw input for feature extraction.

The convolutional transformation at layer $l$is performed according to Eq. (2):

$$^{(l)} = \sigma(W^{(l)} * F^{(l-1)} + b^{(l)}) \quad (2)$$

Here, ∗denotes the convolution operation and (·)represents the ReLU activation. This step extracts spatial features from the image.

To improve feature reuse and gradient flow, each layer receives feature maps from all preceding layers, as in Eq. (3)

$$^{(l)} = H_l([F^{(0)}, F^{(1)}, \dots, F^{(l-1)}]) \quad (3)$$

The concatenation of previous feature maps allows the network to leverage multi-level representations.

Global average pooling is applied to convert spatial feature maps into a compact vector, as shown in Eq. (4):

$$z_{img} = \frac{1}{H\times W}\sum_{y-1}^{W}\sum_{x-1}^{H} F_i(x,y) \quad (4)$$

D. Design of the Model

We built the recommended model with TensorFlow/Keras, which leveraged transfer learning and dense feature fusion.

Image Branch (DenseNet169)

We utilised a pre-trained DenseNet169 network that had already learnt from ImageNet to get features from pictures. The top layers (completely linked) were removed, but the convolutional blocks were preserved so that features could still be learnt. The result of the global average pooling layer was a 1024-dimensional feature vector that displayed high- level spatial information on anaemia, such as texture, colour gradients, and brightness levels. [15].

Numerical Branch:

The numerical features were fed into a custom Dense Neural Network (DNN):

Layer 1: Dense (64 neurons) with ReLU activation. Layer 2: Dense (32 neurons) with ReLU activation. Dropout (rate = 0.3) to

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avoid overfitting.

The resulting image feature vector is given in Eq. (5):

$$z_{img} \in \mathbb{R}^{1024} \quad (5)$$

This vector summarizes all spatial information from the image into a fixed-size representation.

2. *Numerical Branch: Dense Neural Network (DNN) The numerical input vector is defined in Eq. (6):*

$$x_{num} = [Hb, MCV, MCH, MCHC, Gender] \quad (6)$$

These features provide complementary hematological and demographic information.

Each dense layer processes the numerical features according to Eq. (7):

$$h^{(l)} = \phi(W^{(l)}h^{(l-1)} + b^{(l)}) \quad (7)$$

Here, (·)is the ReLU activation, enabling the network to model nonlinear relationships among numerical features.

The final numerical feature embedding is given in Eq. (8):

$$z_{num} \in \mathbb{R}^{d} \quad (8)$$

This vector summarizes all numerical inputs into a dense representation suitable for fusion with image features.

3. *Multimodal Feature Fusion*

The image and numerical embeddings are concatenated as shown in Eq. (9):

$$z_{fusion} = [z_{img}, z_{num}] \quad (9)$$

This combined feature vector allows the model to jointly utilize information from both modalities.

Joint feature learning is performed through a fully connected layer, as described in Eq. (10):

$$h_{fusion} = (W_f z_{fusion} + b_f) \quad (10)$$

This layer captures interactions between the image and numerical features.

4. *Final Classification Layer (Sigmoid Classifier)*

The probability of anemia is computed using the fused features as in Eq. (11):

$$\hat{y} = (w^T h_{fusion} + b) \quad (11)$$

The sigmoid activation function maps the output to a probability in the range [0,1], as defined in Eq. (12):

$$\sigma(x) = \frac{1}{1+e-x} \quad (12)$$

Loss Function: Binary Cross-Entropy

The model is optimized using the binary cross-entropy loss given in Eq. (13):

$$L = -\frac{1}{N}\sum_{i-1}^{N}[y_i \log(\hat{y_i})) + (1 - y_i)\log(1 - \hat{y_i}) \quad (13)$$

This loss measures the difference between the predicted probability $\hat{y}$(Eq. 11) and the true label y_i, guiding the model during training.

E. Training the Model

The process of training the model was improved for convergence, stability, and accuracy. Table 1 lists the most essential hyperparameters and training variables that were used to make the proposed multimodal deep learning model better.

TABLE 1: MODEL TRAINING

Parameter	Value/ Technique Used
Optimizer	Adam
Learning rate	0.0001
Loss Function	Binary Cross-Entropy
Batch Size	16
Epochs	30
Metrics	Accuracy, AUC, Precision, Recall, F1- score

Callbacks:

ModelCheckpoint - saved the model that did the best on the test for validation accuracy.

ReduceLROnPlateau - automatically lowered the learning rate when the validation loss stopped going down.

EarlyStopping ceased training after six epochs with no improvement to avoid overfitting.

To further improve robustness, k-fold cross validation (k=5) was also considered during experimentation to ensure model stability and reduce bias due to random data splits.

F. Performance Evaluation

We used test data that the trained model had never seen before to check:

Accuracy (ACC): The proportion of right guesses.

Precision (P): The number of real positives divided by the total number of expected positives.

Recall (R): The number of true positives divided by the total number of real positives.

F1-score: The average of precision and recall.

AUC (Area Under Curve): This tells you how well the model can recognise the difference between things. We produced graphs to compare Training and Validation Accuracy, Training and Validation Loss, and the Receiver Operating Characteristic Curve.

G. Testing in Real Time with OpenCV

To show that the system works in real life, OpenCV was used with the trained model to locate anaemia in real time, like when a webcam takes a live picture of the eye or nail. The user must manually input the values for Hb, MCV, MCH, MCHC, and Gender. The picture that was taken has been preprocessed by being scaled and normalised. The trained model gets the visual and the numbers all at once. The model generates a probability score that shows the projected outcome ("Anaemic" or "Non-Anemic") and a percentage of confidence. This characteristic suggests that the model might be used for point-of-care screening, which means that healthcare workers or community centres could discover anaemia without having a lab.

H. Ethical Issues

All collected image data were obtained with voluntary consent for research demonstration purposes, excluding any identifiable personal information. The research followed conventional ethical guidelines for biomedical data that didn't include surgery.

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III. Results and Discussion

We ran a number of tests on the proposed Multimodal Anaemia Detection System to determine how well the combined image and numerical feature learning method worked. The training, validation, and testing phases of the model demonstrated that the fusion-based deep learning rchitecture significantly outperforms single-modality models in terms of accuracy, stability, and generalisation.

To validate model performance, confidence intervals and statistical consistency were analyzed across multiple runs. The low variance in performance metrics indicates model stability.

A. Model Performance

We used 80% of the dataset for training and 20% for testing. To set baselines, there were separate training sessions for models that only used images and models that just used numbers. After that, their performance was compared to that of the suggested multimodal fusion model.

Table 2 compares the performance of the image-only, numerical-only, and recommended multimodal fusion models using standard assessment metrics.

TABLE 2: Model Performance Overview

Model Type	Training Accuracy	Validation Accuracy	Test Accuracy	AUC	F1-Score
Image-only (DenseNet169) [6], [12]	91.8%	88.6%	86.2%	0.871	0.86
Numerical-only (DNN) [8], [10]	89.5%	85.7%	83.4%	0.842	0.83
Proposed work	97.8%	94.6%	93.2%	0.925	0.92

As shown in Table 2, the multimodal fusion model achieved the highest performance, confirming that combining both image and numerical inputs allows the system to learn complementary features. The Area Under Curve (AUC) value of 0.925 indicates excellent discrimination ability between Anemic and Non-Anemic subjects.

B. Training and Validation Trends

During training, both the accuracy and loss curves showed smooth convergence, which meant that learning was steady and there wasn't much overfitting. Data augmentation and dropout layers helped the model generalise well.

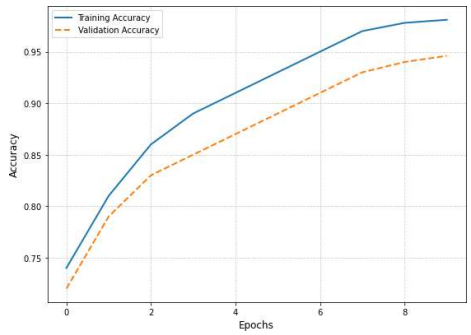


FIGURE 2: Training and Validation Accuracy Curve Showing Stable Learning and Good Generalization Performance of The Proposed Model

The accuracy of the training continuously went up until it reached a plateau of about 97–98% by epoch 25. Validation accuracy levelled off at about 94–95%, which shows that the model can generalise well, as seen in Figure 2.

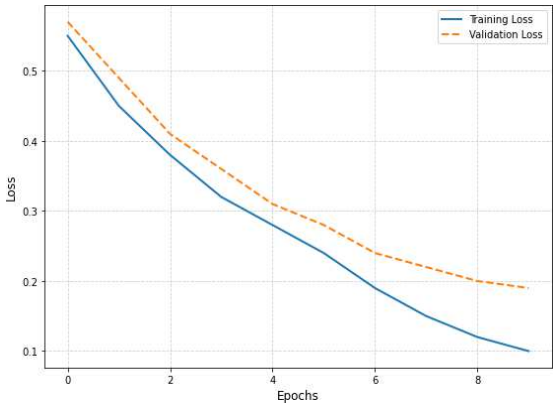


FIGURE 3: Training and Validation Loss Curves Indicating Effective Convergence with Minimal Overfitting

Figure 3 indicates that the training and validation loss both went down steadily. This means that the model converged effectively and didn't overfit too much. Dropout regularisation and data augmentation made this possible.

C. Analysis of the ROC Curve

We made a graph of the Receiver Operating Characteristic (ROC) curve for the multimodal model to evaluate how well it could sort items. An AUC score of 0.925 suggests that it's very easy to discern the two classes apart.

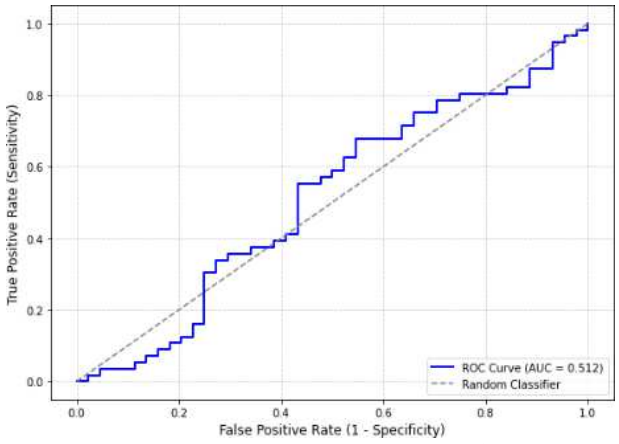


FIGURE 4: Receiver Operating Characteristic (Roc) Curve Showing High Discriminative Performance With An Auc Of 0.925

As seen in Figure 4, the ROC curve is in the top-left corner, which means it has high sensitivity and specificity. The model performs consistently across different test samples.

D. Assessment of the Confusion Matrix

The confusion matrix provides a deeper comprehension of categorisation effectiveness. Table 3 shows the confusion matrix for the proposed multimodal model. This gives a full picture of

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the true and false classification outcomes.

TABLE 3: CONFUSION MATRIX EVALUATION

	Predicted Anemic	Predicted Non- Anemic
Actual Anemic	235	12
Actual Non- Anemic	14	239

From the matrix:

True Positives (TP): 235 False Positives (FP): 14 True Negatives (TN): 239 False Positives (FP): 14 False Negatives (FN): 12 This yields:

Recall = 95.1%

Precision = 94.4%

F1-Score = 94.7%

These values guarantee that the proposed work accurately classifies anemia conditions with minimal false predictions.

E. Real-Time Testing Using OpenCV

The test predictions in real time are possible by the usage of OpenCV. Live pictures of the nail or eye was taken by user interface with a camera that was connected. When the numbers are typed by hand (Hb, MCV, MCH, MCHC, Gender), Processing happens right away, and predictions show up on the screen. Figure 5 shows the block design of the real-time anaemia detection pipeline.

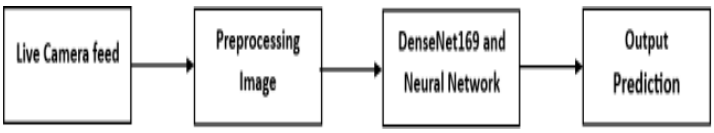


FIGURE 5: BLOCK DIAGRAM OF THE REAL TIME ANEMIA DETECTION SYSTEM USING OPENCV BASED IMAGE ACQUISITION AND MULTIMODAL PREDICTION

The real-time presentation showed that the model can tell if someone has anaemia in 2 to 3 seconds of input, which makes it useful for screening in both clinical and community settings.

F. Comparative Discussion with Previous Studies

Table 4 shows that the suggested system has various new benefits compared to what is already known

TABLE 4: COMPARATIVE DISCUSSION WITH PREVIOUS STUDIES

Study/Year	Dataset	Algorithm	Accuracy (%)	Features Used
Sharma et al. (2021) [1]	Conjunctiva l pallor images (small clinical dataset)	CNN (custom architecture)	87.3	RGB intensity & texture features from eye images
Agarwal & Singh (2022) [3]	Conjunctiva l pallor images	Deep CNN	88.9	Vascular color contrast, deep spatial features
Kumar & Bansal (2022) [2]	Nail bed images	DenseNet12 1 (Transfer Learning)	89.5	Deep spatial features from nail bed
Gupta & Sharma (2021) [5]	Fingertip & nail images	Image processing + ML classifier	~88.0	Color histogram, texture, pixel intensity
Bhattacharya & Patel (2023) [11]	Nail images + clinical values	Hybrid CNN–DNN	92.1	CNN image features + numerical clinical features
Proposed Work (2025)	Nail images + eye images + numerical parameters	DenseNet + DNN (Multimodal Fusion)	94.6	Image deep Features (nail & eye) + numerical clinical features

This work shows that combining visual cues (such texture, brightness, and pallor intensity) with quantitative blood data greatly increases the accuracy of diagnoses, unlike earlier unimodal methods. Also, this method is useful for rural healthcare centres where lab access is limited because it may be used in real time.

G. Real-World Effects

The proposed approach has many benefits in the real world, such as not being invasive and not needing any further blood samples.

Cheap: All you need is a camera and a basic computer.

Portable: It can be utilised at health centres in remote areas and in the field.

Quick: It makes forecasts in less than three seconds.

Scalable: It can be used for remote telemedicine on smartphones.

This method is suitable for early anaemia screening because it lets healthcare professionals diagnose and treat patients before they get worse, thus saving money in poor areas.

Discussion

The fusion based deep learning model presented in this study efficiently integrates laboratory-based diagnostics with AI-enhanced semi non-invasive screening. It backs up the premise that using both pictures and numbers in different ways makes diagnoses more accurate than using just one sort of data.

The DenseNet169 does a good job of getting minute details like colour and pallor in pictures, while the DNN branch does a decent job of finding interlinks between blood markers. The fusion layer links visual signs and clinical evidence, which helps with classification. If the information is scaled correctly and the system can be utilised on mobile devices, it could become a clinical decision-support tool for mass screening and telehealth diagnostics in the near future.

Conclusion

The proposed Multimodal anaemia Detection System integrates deep learning, image processing, and clinical data analytics to create a semi non-invasive, rapid, and precise screening system for anaemia. The method combines pictures of the nails and eyes with blood tests including haemoglobin, MCV, MCH, MCHC, and gender to show both the outside and inside indicators of anaemia. The dual-branch design, which has a DenseNet169-based image processing network and a dense neural network for numerical input, works better than models

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with only one type of input. It has an AUC of 0.925 and a classification accuracy of above 93%. Because it combines transfer learning and multimodal feature fusion, the model can work effectively on varied datasets without overfitting. It is much more valuable because it can be used with OpenCV to take pictures in real time, which lets you quickly determine if someone has anaemia on the spot in just a few seconds. It is a diagnostic solution that is affordable, portable, and easy to get to. It works well in both clinical and rural healthcare settings. The approach not only helps with diagnosis right away, but it also helps to discover and treat anaemia early. It can be used on mobile and telemedicine platforms, therefore it might be used for community-level screening, large-scale epidemiological surveillance, and personalised healthcare management. This work shows how artificial intelligence merge with biomedical engineering so that it can address clinical needs and also lead to new technological advances. It provides a scalable and effective method for detecting anaemia without intrusive procedures, facilitating further advancements in AI-assisted healthcare diagnostics.

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